EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY

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NAAC GRADE A

**Department of Computer Science & Engineering**

# CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

## HALESH HL

bearing USN 1EP24CS050, Section A, has successfully completed the Advanced UI/UX activity (Problem No. 20) titled

**“Metro Control Room Operator Console”**

& has submitted this report as required on 11-11-2026, during the academic year 2026-27.

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# Abstract

The Metro Control Room Operator Console is a web-based user-interface prototype intended to help metro control-room operators view essential operational information in one place. The project addresses the challenge of switching between separate views while monitoring trains, responding to technical faults, coordinating passenger incidents and completing shift activities. The primary users considered are control-room operators, with station and response personnel as supporting stakeholders. The solution brings together an overview dashboard, train monitoring, fault alerts, passenger incident management and shift management. It is implemented as a front-end prototype using HTML5 for structure, CSS3 for presentation and JavaScript for interactive behaviour. Demonstration records are used to show train status, alert acknowledgement and resolution, incident creation, assistance selection and simulated dispatch. The interface uses a dark control-room visual style, structured panels, status indicators and consistent navigation to support scanning and task completion. Functional checks carried out during development confirmed that the principal screens and simulated workflows operate in the local browser. The prototype is not connected to live metro systems and does not contact emergency services; real-user research and formal usability validation remain future work.

# 1. Problem Definition and Scope

**Problem statement**

Metro operators must maintain awareness of train movements, technical faults, passenger incidents and shift handovers. When information is scattered across separate tools or presented without clear priority, operators may spend extra time locating details and coordinating a response. This project explores a centralized console that groups these tasks into a consistent interface.

**Stakeholders and target users**

Primary users: metro control-room operators. Supporting stakeholders: station staff, train operations personnel, maintenance teams, passenger-assistance teams, medical responders and shift supervisors. These are intended stakeholder groups for the design; no specific organization or live operating procedure is represented.

**Goals and success metrics**

Goals are to make key operational sections easy to locate, show train and alert status clearly, support a traceable incident workflow and present shift handover information. Prototype-level success checks include correct rendering of records, working navigation, correct status changes after actions and clear feedback after simulated dispatch.

**Constraints and assumptions**

The prototype uses static or locally simulated data and browser-side JavaScript. It assumes a desktop control-room display and an operator familiar with basic computer use. It has no live train feed, backend database, authentication, dispatch integration or verified railway safety certification.

**Scope: what the design will and will not cover**

Included: overview, train monitoring, fault alerts, passenger incident reporting and tracking, assistance selection, simulated dispatch and shift management. Excluded: real train control, signalling, timetable optimization, passenger identity records, live emergency calls, production security and integration with metro infrastructure.

# 2. User Research

**Research method and questions asked**

The project brief and prototype requirements were used to identify likely operator tasks and information needs. No completed interview transcripts, survey responses or observation records were supplied for this report; therefore this section records a research plan and design assumptions, not claimed fieldwork. Suggested questions: What information must remain visible during a shift? How are delays and faults prioritized? What details are needed when reporting a passenger incident? How is assistance requested and tracked? What must be included in a shift handover?

**Participant details**

The intended research group is operators or people familiar with transport operations, station personnel and users who coordinate passenger assistance. A minimum of five target-group participants is required by the template. Participant names, roles, consent and interview dates must be entered only after actual sessions are completed.

**Key findings and direct observations**

Validated direct observations are not available in the supplied project material. The design assumptions are that operators benefit from a consolidated overview, prominent status labels, concise train details, prioritized alerts and a clear incident progression. These assumptions should be checked with participants.

**Competitor or existing solution analysis**

The prototype is not based on a documented audit of a named commercial metro control system. A general comparison of common dashboard patterns suggests useful practices: persistent navigation, clear severity levels, tabular operational records, filters, acknowledgement actions and activity status. A formal competitor study should cite the systems actually reviewed.

**Insights that will drive the design**

Keep high-priority information easy to scan; separate train, fault, incident and shift tasks; show status in text as well as colour; make incident assistance choices explicit; provide feedback after actions; and distinguish simulated actions from real dispatch.

| No. | Participant profile | Method | Key observation |
|---|---|---|---|
| 1 | To be completed | Interview | Record actual observation |
| 2 | To be completed | Interview | Record actual observation |
| 3 | To be completed | Interview | Record actual observation |
| 4 | To be completed | Interview | Record actual observation |
| 5 | To be completed | Interview | Record actual observation |

Research evidence status: participant rows are intentionally left for genuine data; do not submit them as completed interviews.

# 3. Personas and Empathy Map

**Persona 1 – Control-room operator (provisional)**
Role: operator responsible for monitoring network activity. Goals: understand current train status, identify abnormal conditions quickly, acknowledge faults and coordinate passenger assistance. Needs: concise records, clear severity, consistent controls and reliable feedback. Frustrations to validate: information scattered across screens, ambiguous statuses and repeated data entry. This is a provisional design persona, not an interviewed individual.

**Persona 2 – Station / passenger-assistance coordinator (provisional)**
Role: staff member who receives or supports passenger incident requests. Goals: understand incident location and type, identify the assistance required and communicate progress to the control room. Needs: clear incident details, visible assignment and progress state. Frustrations to validate: incomplete incident details and uncertainty about whether help has been dispatched. This is a role-based persona, not a real participant.

**Empathy map: says, thinks, does, feels**
Says: “I need the current status and location.” Thinks: “Which issue needs attention first?” Does: scans the overview, checks train records, acknowledges alerts and updates incident status. Feels: may feel time pressure when several events occur together; a clear hierarchy and feedback can reduce uncertainty. These are hypotheses for validation, not direct quotations.

**Pain points and unmet needs**
Potential pain points include switching between operational categories, difficulty distinguishing active from resolved issues, incomplete incident information and unclear handover status. Unmet needs include a consolidated view, readable status labels, simple response actions and a history of what has been done.

**Problem reframed as a design question**
How might we provide metro control-room operators with a clear, centralized and easy-to-scan interface that helps them monitor trains, manage faults, coordinate passenger incidents and hand over shift information with fewer navigation steps?

# 4. User Journey and Information Architecture

**End-to-end user journey with emotional highs and lows**

1. Open console and scan overview (orientation). 2. Notice a train delay or fault (attention rises). 3. Open Train Monitoring or Fault Alerts and inspect details (understanding). 4. Acknowledge or resolve the alert after the required action (progress feedback). 5. Open Passenger Incidents, record the event and select assistance (coordination). 6. Simulate dispatch and update the incident state (confirmation). 7. Review shift information and handover items (closure). Potential stress points are multiple simultaneous alerts and incomplete incident details; clear hierarchy and status feedback are intended to help.

**Task flows for the three most critical tasks**

Train review: Overview → Train Monitoring → select/read train record → inspect route, speed and status. Fault response: Overview → Fault Alerts → inspect severity and description → acknowledge → resolve after action. Passenger incident: Passenger Incidents → review or report incident → choose assistance type → dispatch (simulation only) → mark In Progress → mark Resolved.

**Sitemap or navigation structure**

MetroOps Console → Overview; Train Monitoring; Fault Alerts; Passenger Incidents; Shift Management. The overview provides the high-level operational summary, while each functional section contains the detailed records and actions for that task.

**Content and feature prioritisation**

High priority: network summary, active train status, critical fault visibility, incident details and assistance workflow. Medium priority: alert acknowledgement history, shift details and handover checklist. Future priority: search, filters, audit trail, role-based permissions, live data and reporting analytics.

# 5. Ideation and Concept Selection

**Rough sketches or rapid ideas**

Early layout exploration can be represented as three alternatives: (A) a single-page dashboard with summary cards and detailed sections; (B) separate pages for each operational task; (C) a map-first interface with contextual side panels. No scanned paper sketches were supplied with the project, so this report describes the alternatives rather than claiming scanned sketches are attached.

**At least three distinct concepts**

Concept A – Unified dashboard: summary at the top and detailed operational sections below. Concept B – Tabbed workspace: one main task visible at a time with navigation tabs. Concept C – Map-centred console: network map dominates and records open in contextual panels.

**Evaluation of concepts against criteria**

The concepts were considered against visibility, navigation effort, information density, ease of implementation and suitability for a desktop control-room workflow. A unified dashboard offers strong at-a-glance visibility; tabs reduce page length but can hide other sections; a map-centred concept offers spatial context but needs accurate live map data.

**Chosen concept and reason for selection**

Concept A was selected for the current prototype because it keeps the major operational areas accessible in one continuous dashboard and makes it easier to compare the overview with detailed records. It also fits the implemented HTML, CSS and JavaScript prototype.

| Criteria | Concept A | Concept B | Concept C |
|---|---|---|---|
| At-a-glance visibility | High | Medium | High |
| Navigation effort | Low | Medium | Medium |
| Information density | High | Medium | Medium |
| Implementation effort | Medium | Low | High |
| Live map dependency | Low | Low | High |

# 6. Low-Fidelity Wireframes

**Wireframes of key screens**

The low-fidelity structure is represented by the implemented screen hierarchy: (1) Overview with operational summary and network information; (2) Train Monitoring with train records; (3) Fault Alerts with alert details and actions; (4) Passenger Incidents with incident records and assistance controls; (5) Shift Management with operator and handover information. Separate scanned wireframe images were not included in the supplied materials.

**Annotations explaining layout and interaction decisions**

The overview is placed first to establish context. Detailed records are grouped by task to reduce mixing of train, fault and passenger information. Status and severity are shown near the relevant record. Action controls are placed beside the item they affect. Incident assistance is selected before the simulated dispatch action.

**Quick feedback from at least two people**

No named or dated feedback records were provided. The template requires feedback from at least two people; record the participant role, task, comment and date after collecting genuine feedback. Do not treat developer self-review as external user feedback.

**Changes made after feedback**

During development, the dashboard was adjusted so the main operational sections remain visible in the overview flow. A JavaScript initialization issue that stopped register data from rendering was corrected by checking for an optional refresh-button element before attaching its event listener. This is a development fix, not evidence of external user feedback.

# 7. Visual Design and Design System

**Colour palette with contrast ratios**

The prototype uses a dark navy/near-black control-room background with blue and teal accents, neutral text and status colours. Exact HEX values and measured contrast ratios were not supplied; therefore no numerical WCAG contrast claim is made. Before final accessibility sign-off, record the actual CSS colour values and verify text and control contrast using a contrast analyser.

**Typography and type scale**

A clear sans-serif interface typeface is used for screen readability. Use a consistent hierarchy: page title, section heading, card label, body text and compact metadata. Maintain readable body sizing and avoid relying on very small text for critical train or incident information.

**Spacing, grid and layout rules**

Use a consistent page gutter, aligned cards, predictable vertical spacing and grouped information panels. Keep related labels and values together. Use responsive wrapping where the viewport narrows, while prioritising the desktop control-room layout.

**Reusable components and their states**

Reusable patterns include navigation items, KPI cards, status badges, train rows, alert panels, incident cards, action buttons and checklist items. Relevant states include normal, selected, active, acknowledged, in progress, resolved, disabled and empty.

**Iconography and imagery choices**

Simple interface icons and restrained decorative elements support quick recognition. Icons should accompany text labels for critical actions so meaning is not communicated by icon alone. The project does not require decorative photography for its operational data views.

**Rationale linking visual choices to user needs**

The dark interface is intended to suit a control-room visual environment and make status accents noticeable. Consistent grouping supports scanning; text labels alongside colour improve clarity; repeated components reduce the effort needed to learn each section.

# 8. High-Fidelity Interactive Prototype

**Final screens for all critical flows**

The implemented prototype includes Overview Dashboard, Train Monitoring, Fault Alerts, Passenger Incident Management and Shift Management. Train records show identifiers and operational details; fault records show alert information and acknowledgement/resolution actions; incident records support reporting, assistance selection and progress tracking.

**Interactions, transitions and micro-interactions**

JavaScript handles navigation and simulated state updates. Operators can inspect train records, acknowledge and resolve fault alerts, create or review passenger incidents, choose First Aid Team, Ambulance, Medical Doctor or Emergency Response Team, simulate dispatch, update incident progress and use shift handover controls.

**Edge states: empty, loading, error, offline and permission denied**

The current prototype primarily demonstrates locally available sample data. A production version should define explicit empty, loading, error, offline and permission-denied states. The present prototype does not implement real authentication, network connectivity checks or role-based permissions.

**Prototype link and access instructions**

Repository: https://github.com/halesh1/Metro-Control-Room-Operator-Console. Open the Prototype/Prototype folder, open index.html in Visual Studio Code and run it using Live Server. The GitHub repository contains the HTML, CSS and JavaScript source files.

**Tools and resources used**

HTML5, CSS3 and JavaScript; Visual Studio Code for development; Google Chrome for local browser checks; GitHub for source control and project hosting. The prototype uses simulated data and does not connect to a real metro operations platform.

# 9. Usability Testing and Iteration

**Test plan: tasks, participants and success criteria**

The local prototype was checked during development for screen loading, navigation, record rendering and action behaviour. The template additionally calls for testing with at least five target users. No five-person test log, participant consent record or measured task times were supplied; therefore a formal five-user usability study is not claimed. Suggested tasks: locate a delayed train; acknowledge a fault; report a passenger incident; dispatch simulated assistance; complete a handover checklist.

**Results: task success, time on task, errors and satisfaction**

Development checks confirmed that the main dashboard sections and simulated interactions worked in the local browser. Quantitative task-success rates, completion times and satisfaction scores were not collected in the supplied evidence and are therefore not reported.

**Accessibility check and issues fixed**

A preliminary design review should verify keyboard access, visible focus, semantic headings, form labels, status text, contrast and screen-reader announcements. WCAG 2.2 AA conformance has not been formally audited. Status should not rely on colour alone; controls should have clear accessible names.

**Problems found, ranked by severity**

A high-impact development issue occurred when JavaScript attempted to attach a listener to an optional overview refresh element that was absent; execution stopped before records rendered. The listener was guarded so initialization can continue. Further formal severity ranking requires user testing.

**Design changes made and evidence of improvement**

The optional refresh control is now checked before attaching its listener. The train, alert and incident rendering functions can then run during initialization. Local functional checks showed the operational registers populated after the correction. Evidence is development verification, not measured user-study improvement.

| User | Task | Success (Y/N) | Time | Issues observed |
|---|---|---|---|---|
| Developer check | Load dashboard | Yes | Not timed | None recorded |
| Developer check | View train records | Yes | Not timed | None recorded |
| Developer check | Acknowledge alert | Yes | Not timed | None recorded |
| Developer check | Dispatch simulation | Yes | Not timed | Simulation only |
| User study | To be completed after actual testing | — | — | Collect actual data |

*Note: Developer checks are not five independent user participants. Complete a separate five-user study and enter actual findings before final academic submission.*

# 10. Design Rationale, Reflection and Future Work

**Major design decisions and trade-offs**

A unified dashboard was chosen to keep operational categories accessible without repeatedly changing pages. This improves overview visibility but creates a longer page that requires scrolling. Detailed sections remain grouped to avoid mixing unrelated records. Simulated actions make the workflow demonstrable without implying a live emergency connection.

**Limitations of the final design**

The prototype has no live train feed, backend persistence, operator authentication, role-based access, audit logging or integration with signalling and emergency services. Data is illustrative. It has not undergone formal five-user usability testing, a complete WCAG 2.2 AA audit or operational safety validation.

**What failed or was redone, and what was learned**

During development, register data did not appear because a JavaScript event listener was attached to a missing optional element. Guarding the listener prevented that initialization error. The issue highlighted the importance of checking DOM elements, validating scripts and testing every major section after changes.

**Ethical considerations**

Operational and passenger data in a real system would be sensitive. A deployed solution must minimise personal data, restrict access by role, protect records, provide audit trails and follow applicable privacy and safety requirements. This demonstration uses sample data and does not dispatch real emergency services.

**Future improvements**

Connect to approved real-time data sources; add secure authentication and role permissions; persist records in a database; provide searchable audit history; add accessible notifications and keyboard support; conduct interviews and five-user usability tests; measure task completion; complete WCAG 2.2 AA review; and validate workflows with qualified metro operations stakeholders.

# References and Appendix

**References**

[1] World Wide Web Consortium (W3C), “Web Content Accessibility Guidelines (WCAG) 2.2,” W3C Recommendation, Oct. 2023. https://www.w3.org/TR/WCAG22/ (accessed Nov. 11, 2026).
[2] MDN Web Docs, “HTML: HyperText Markup Language.” https://developer.mozilla.org/en-US/docs/Web/HTML (accessed Nov. 11, 2026).
[3] MDN Web Docs, “CSS: Cascading Style Sheets.” https://developer.mozilla.org/en-US/docs/Web/CSS (accessed Nov. 11, 2026).
[4] MDN Web Docs, “JavaScript.” https://developer.mozilla.org/en-US/docs/Web/JavaScript (accessed Nov. 11, 2026).
[5] Metro Control Room Operator Console, project source code and documentation, GitHub repository. https://github.com/halesh1/Metro-Control-Room-Operator-Console (accessed Nov. 11, 2026).

**Appendix A – Research records**

Attach actual interview scripts, participant consent notes, raw notes and survey responses after collection. No participant records were supplied with the project materials used to prepare this report.

**Appendix B – Usability test records**

Attach task scripts, participant details, task outcomes, time measurements, observed errors and satisfaction responses after conducting the required usability sessions.

**Appendix C – Additional design evidence**

Recommended evidence to attach: screenshots of Overview, Train Monitoring, Fault Alerts, Passenger Incidents and Shift Management; early sketches or wireframes; final prototype screenshots; and any actual feedback-driven revisions.

**Submission note**

This report documents the current front-end prototype and distinguishes implemented features from research, testing and accessibility work that still requires evidence. The assistance dispatch workflow is simulated and does not contact real emergency services.